

Q₂³

$$\dot{x} = Ax + Bu \quad y = Cx + Du \quad \left. \vphantom{\dot{x} = Ax + Bu} \right\} \text{state equations}$$

$$\hat{A} = TAT^{-1} \quad \hat{B} = TB \quad \hat{C} = CT^{-1} \quad \hat{D} = D$$

→ equivalent state space representation of the system.

$$\hat{x} = Tx$$

→ state transformation

Controllable canonical form:-

$$H(s) = \frac{P_{n-1}s^{n-1} + \dots + P_1s + P_0}{s^n + \alpha_{n-1}s^{n-1} + \dots + \alpha_1s + \alpha_0} + d_0$$

$$A_{ccf} = \begin{bmatrix} 0 & 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \dots & 1 \\ -\alpha_0 & -\alpha_1 & -\alpha_2 & -\alpha_3 & \dots & -\alpha_{n-1} \end{bmatrix}$$

$$B_{ccf} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}^T \quad C_{ccf} = [P_0 \ P_1 \ \dots \ P_{n-1}]^T$$

$D_{ccf} = d_0$

$$\frac{y(s)}{U(s)} = \frac{P_{n-1}s^{n-1} + \dots + P_1s + P_0}{s^n + \alpha_{n-1}s^{n-1} + \dots + \alpha_1s + \alpha_0} + d_0 \Rightarrow y = \sum_{i=1}^n \frac{P_i s^{i-1}}{\sum_{j=0}^n \alpha_j s^j} u + d_0 u$$

$$x_i = \frac{s^{i-1}}{\sum_{j=0}^n \alpha_j s^j}$$

$$x_{i+1} = \frac{s^i}{\sum_{j=0}^n \alpha_j s^j} \Rightarrow \dot{x}_i(t) = \frac{dx_i}{dt}$$

$$x_2 = \dot{x}_1, x_3 = \dot{x}_2, \dots, x_n = \dot{x}_{n-1}$$

Observable canonical form:-

$$A_{ocf} = A_{ccf}^T \quad B_{ocf} = B_{ccf}^T$$

$$A_{ocf} = \begin{bmatrix} 0 & 0 & 0 & \dots & 0 & -\alpha_{n-1} \\ 1 & 0 & 0 & \dots & 0 & -\alpha_{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & -\alpha_1 \end{bmatrix}$$

$$C_{ocf} = C_{ccf}^T \quad D_{ocf} = D_{ccf}$$

$$B_{ocf} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

$$C_{ocf} = [P_0 \ P_1 \ \dots \ P_{n-1}]$$

$$D_{ocf} = d_0$$

$$y = \beta_0 x_1 + \beta_1 x_2 + \dots + \beta_{n-1} x_n + d_0 u$$

$$\frac{y}{u} = \frac{\beta_{n-1}s^{n-1} + \dots + \beta_1 s + \beta_0}{s^n + \alpha_{n-1}s^{n-1} + \dots + \alpha_1 s + \alpha_0} + d_0 \quad \Rightarrow \quad \beta_{i-1} x_i = \frac{\beta_{i-1} s^{i-1}}{s^n + \alpha_{n-1}s^{n-1} + \dots + \alpha_1 s + \alpha_0}$$

$$\Rightarrow x_i = \frac{s^{i-1}}{s^n + \alpha_{n-1}s^{n-1} + \dots + \alpha_1 s + \alpha_0}$$

$\Rightarrow$ State variables same as controllable canonical form.

Diagonal canonical form:-

$$H(s) = b_0 + \frac{C_1}{s+p_1} + \frac{C_2}{s+p_2} + \dots + \frac{C_n}{s+p_n}$$

$$A_{def} = \begin{bmatrix} -p_1 & 0 & \dots & 0 \\ 0 & -p_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & -p_n \end{bmatrix}$$

$$B_{def} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$$

$$C_{def} = [C_1 \ C_2 \ \dots \ C_n]$$

$$D_{def} = b_0$$

$$y = C_{def} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + D_{def} u$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_n \end{bmatrix} = A_{def} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + B_{def} u$$

$$\frac{y(s)}{u(s)} = b_0 + \frac{C_1}{s+p_1} + \frac{C_2}{s+p_2} + \dots + \frac{C_n}{s+p_n}$$

$$y(s) = b_0 u + \frac{C_1 u}{s+p_1} + \frac{C_2 u}{s+p_2} + \dots + \frac{C_n u}{s+p_n}, \text{ since } y = Cx + Du$$

$$x = \begin{bmatrix} \frac{u}{s+p_1} \\ \frac{u}{s+p_2} \\ \vdots \\ \frac{u}{s+p_n} \end{bmatrix}$$

$\Rightarrow$

State variables different from CCF,
DCF